

Polymers and Polymer Processing

Feature	Thermoplastic Polymers	Thermosetting Polymers
Structure	Predominantly linear with a degree of branching and side chains	Polymerise to form a three-dimensional network which cannot be melted. Cross-linked 3D network after curing
Processing	Can be Melted and caused to flow/reshaped repeatedly	Irreversibly set during curing (can't be remelted)
Thermal Properties	Generally poor (soften at moderate heat)	Better temperature resistance
Initial Form	Supplied as solid pellets -Typically supplied as pellets ready for conversion via a thermal process to form the shape required	Often liquid or paste, cured in mould. often in the form of a liquid or solution that can be moulded. Transforms from the initial liquid (which is typically a partially reacted polymer) into a macromolecule in a mould. This process often involves some form of cross-linking reaction.
Moulding Process	Heated to soften, then shaped	Cured via cross-linking (e.g., polymer + solvent reaction). – for example, unsaturated polyesters actually react with styrene (the solvent) to form a co-polymer of polyesterpolystyrene.
Mechanical Properties	More flexible, lower strength	Relatively hard , more rigid (for a polymer)
Examples	Nylon, Polyethylene, ABS, PET, Polypropylene	Epoxy resin, Unsaturated polyester
Recyclability	Recyclable (can be remelted)	Not recyclable via remelting

Shape Change Process in Polymers

Polymers are usually processed using a moulding operation involving shape changes in a liquid state.

At much lower temperatures than those required for metals.

At lower pressures and hence lower tooling costs when compared to metals.

Moulding of Thermoplastics and Thermosets

The difference in processing between thermoplastics and thermosets is that:

Thermosets solidifies by virtue of a chemical reaction where the liquid transforms to a solid by a process called curing - often requiring heat or a hot mould.

Thermoplastics solidify by cooling in the mould. Thermoplastics therefore generally require cold (or cool) moulds.

At processing temperatures, the viscosity of thermosets is often very low (like water) while that of a thermoplastics is much higher (like treacle)

- **Thermoplastics** → processes that involve **melting and cooling**
- **Thermosets** → processes that involve **curing (chemical reaction + heat)**

Processing Method	Used For	Notes
Injection Moulding	✓ Thermoplastics	Most common method. Pellets melted and injected. Fast & precise.
Gas-Assisted Injection	✓ Thermoplastics	Variant of injection moulding — good for hollow or complex parts .
Co-Injection / Scrim	✓ Thermoplastics	Used to combine multiple polymers in one part.
Extrusion	✓ Thermoplastics	Continuous products (e.g., pipes, sheets, films).
Blow Moulding	✓ Thermoplastics	Bottles and hollow containers. Uses air pressure to form shape.
Vacuum Forming	✓ Thermoplastics	Heated sheets formed over a mould by vacuum. Simple, low-cost shapes.
Blown Film	✓ Thermoplastics	For thin plastic films (e.g., bags, packaging).
Rotational Moulding	✓ Thermoplastics	Good for large hollow shapes like tanks or toys.

Processing Method	Used For	Notes
Compression Moulding	✓ Thermosets	Heated mould + pressure cures polymer into shape.
Resin Transfer Moulding	✓ Thermosets	Resin injected into fibre mat inside mould — used in composites.
Casting (e.g., Epoxy)	✓ Thermosets	Liquid thermoset poured into a mould, cures chemically.
Sheet Moulding Compound	✓ Thermosets	Pre-mixed thermoset paste pressed in moulds (often automotive parts).

Injection Moulding

- This process is designed to facilitate high speed processing of plastics into parts (which may be intricate).
- A feed unit introduces the polymer into the barrel.
- This barrel contains a screw that transports the granules into a chamber.
- The screw does work on the polymer which heats it up and induces melting - a process assisted by additional heating as necessary.

Other Forms of Injection Moulding ● Gas assisted injection moulding, which is used to produce hollow parts. As the gas is injected into the mould cavity. ● Co-injection moulding, is used to mould multiple polymers into a single part. ● Scrim is a moulding process where multiple injection barrels are used with a single tool. Once the injection cycle is completed, the pressure is oscillated before melt freezing occurs. This introduces a preferred and controlled orientation, reducing voids but is expensive and can damage the mould.

Blow Moulding ● Blow moulding was originally developed for glasses and adapted successfully to polymers. ● The material (the parison) is supplied hot in tubular form enclosed by the split mould,

and forced to adopt the shape by air (or gas) pressure. The process is ideal for producing simple shapes, like containers where the surface finish is only critical on the outside. ● The process is cheap as tooling costs are low as the pressures are not high and only small clamping forces required.

Vacuum Forming ● Here the required shape is formed using air pressure. A polymer sheet is drawn as a hot formable sheet onto a tool via the application of a vacuum. ● Again the process is cheap with a low cost tool due to the low pressures. The material requirements for the polymer will be similar to that required for blow moulding. ● The complexity of shape that can be made in this way is obviously limited but large shapes can be made economically.

Blown Film ● An extruder is fitted with an annular die, pointing upwards. ● This tube is blown up with air, and is drawn up in continuous process. The air is contained in the bubble by collapsing rolls at the top. ● Like blow moulding this induces a biaxial orientation, which is then made permanent by the crystallisation process which freezes the orientation in place. ● This process can form either sheets or bags. Bags need to have the edges cut off and require rewinding into two different rolls. The bags require heat sealing at regular intervals and regular perforations to permit tear off or cutting

Rotational Moulding ● Some simple shapes may be produced by rotating a large heated mould with a charge of polymer. ● The polymer inside the mould melts and coats the inside of the mould. ● This will then be allowed to cool whilst still rotating allowing a uniform build up of polymer and ultimately a part which has conformed to the mould tool.